AI-DRIVEN AIR QUALITY MONITORING AND PREDICTION SYSTEM

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA1706-Aritificial Intelligence

to the award of the degree of
BACHELOR OF ENGINEERING
IN

Computer Science of Engineering
Submitted by

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DECLARATION

WE, A. Reddy Prasad (192472045), N. Pramod (192572057) P. Sai Lohith (192311286). Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **"AI-DRIVEN AIR QUALITY MONITORING AND PREDICTION SYSTEM"** the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **"AI-DRIVEN AIR QUALITY MONITORING AND PREDICTION SYSTEM"** has been carried out by A. Reddy Prasad (192472045), **N. Pramod (192572057), P. Sai Lohith (192311286).** under the supervision **Dr. Krishna Kumar** is submitted in partial fulfilment of the requirements for the current semester of the **B. tech-AIML** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Submitted for the Project work Viva-Voce held on _________

INTERNAL EXAMINER EXTERNAL EXAMINER

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TABLE OF CONTENTS

ABSTRACT

Air pollution is a major environmental and public-health concern, particularly in urban and industrial regions. Air quality can change rapidly because of traffic, industrial activity, weather conditions, construction activity and other local factors. Conventional monitoring methods may provide measurements at limited locations and may not offer immediate predictive insight. This project proposes an AI-Driven Air Quality Monitoring and Prediction System that collects air-quality and environmental data, processes the data, calculates or uses Air Quality Index (AQI) information, and applies machine-learning techniques to predict future air-quality conditions.

The proposed system integrates data acquisition, preprocessing, feature engineering, machine-learning prediction and dashboard visualization. Sensor or public environmental data may include particulate matter (PM2.5 and PM10), temperature, humidity and gaseous pollutant measurements where available. After validation and cleaning, the data is transformed into features such as time, date, historical pollutant values and weather conditions. Prediction models such as Linear Regression, Random Forest, Gradient Boosting or time-series models can then be evaluated to estimate future AQI or pollutant concentrations.

The system provides a centralized dashboard for displaying current measurements, historical trends, predicted air quality and alerts. The project demonstrates how artificial intelligence and data analytics can support environmental monitoring and informed decision-making. The proposed platform can be extended with real-time IoT sensors, cloud deployment, automated alerts, geospatial visualization and advanced deep-learning models.

Keywords: Air Quality Monitoring, Artificial Intelligence, AQI Prediction, Machine Learning, IoT, Environmental Analytics, PM2.5, PM10, Data Visualization, Predictive Analytic

CHAPTER 1

INTRODUCTION

1.1 Background Information

Air quality is influenced by particulate matter, gaseous pollutants, traffic emissions, industrial processes and meteorological conditions. Important indicators can include PM2.5, PM10, temperature, humidity and pollutant concentrations such as carbon monoxide or nitrogen dioxide when sensors or datasets are available. Because these factors vary over time, environmental data must be collected and analysed continuously to understand changing conditions.

Artificial intelligence and machine learning provide a practical way to analyse historical environmental patterns and estimate future conditions. By combining monitoring data with time and weather-related features, a prediction model can identify relationships that are difficult to observe through manual analysis alone. The proposed project integrates data collection, preprocessing, prediction and visualization into one decision-support system.

1.2 Project Objectives

- To collect air-quality and environmental data from sensors, APIs or datasets.
- To clean, validate and preprocess environmental measurements.
- To analyse historical pollutant and AQI patterns.
- To develop and evaluate machine-learning models for air-quality prediction.
- To identify periods of poor and improved air quality.
- To provide a dashboard for current, historical and predicted information.
- To support timely awareness and environmental decision-making.
- To design a scalable architecture for future real-time monitoring.

1.3 Significance

The project is significant because reliable environmental information can support awareness, planning and risk reduction. Predictive analytics can move the system beyond displaying only current values by estimating possible future conditions. The system also demonstrates a practical integration of AI, data engineering, IoT-oriented monitoring and visualization.

1.4 Scope

- PM2.5 and PM10 monitoring where data is available.
- Temperature and humidity analysis.
- Optional gaseous pollutant measurements.
- AQI or air-quality category analysis.
- Historical trend analysis and prediction.

- Data visualization and reporting.
- Potential real-time sensor or cloud integration.

The project is intended as a monitoring and decision-support system. Predictions are estimates and should not be treated as guaranteed measurements or as a replacement for certified regulatory monitoring.

1.5 Methodology Overview

The methodology consists of data acquisition, preprocessing, exploratory analysis, feature engineering, model training, evaluation, prediction and dashboard development. Raw measurements are validated for missing values, duplicates, invalid ranges and inconsistent timestamps. Relevant features are then used to train candidate models, which are compared using suitable evaluation metrics. The selected model produces predictions that are presented with historical and current information through the dashboard.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the problem

Air pollution levels can change significantly across time and locations. A system that only displays a current reading provides limited support for anticipating future conditions. Another challenge is that environmental data may come from different sensors or sources with different sampling intervals, units and data quality. The central problem addressed by this project is: How can AI and data analytics be used to collect, process, monitor and predict air-quality conditions in a reliable and understandable manner?

2.2 Evidence of the problem

Environmental monitoring commonly produces time-series data with missing values, measurement noise and changing seasonal patterns. Pollutant levels may be affected by weather, traffic and local activity. These characteristics make data preprocessing and predictive modelling important. A prediction system must therefore be evaluated against actual observations rather than assuming that any single algorithm is universally accurate.

2.3 Stakeholders

- Citizens and communities seeking understandable air-quality information.
- Environmental authorities and municipal organizations.
- Researchers and data analysts.
- Healthcare and public-awareness organizations using aggregated environmental information.
- Industries and institutions monitoring local environmental conditions.
- Engineers and developers maintaining the monitoring platform.

2.4 Supporting Data/Research

The system is supported by the general availability of environmental sensor data and public air-quality datasets. Machine-learning and time-series methods are suitable for modelling relationships between historical measurements, weather features and future pollutant levels. The project uses this established data-driven approach while emphasizing data quality, model validation and transparent communication of uncertainty.

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 Development and design process

Module 1: Data Acquisition. The system receives environmental data from IoT sensors, public datasets or approved APIs. Possible fields include timestamp, location, PM2.5, PM10, temperature, humidity and optional gaseous pollutants.

Module 2: Data Processing. Missing values, duplicates, invalid ranges and inconsistent units are identified before analysis. Timestamps are standardized and data is converted into a structured format.

Module 3: AI Prediction. Historical data is transformed into features such as hour, day, month, previous pollutant values and weather information. Candidate models are trained and compared.

Module 4: Dashboard and Alerts. The dashboard displays current measurements, historical trends, predicted values and air-quality categories. Alerts can be generated when configured thresholds are exceeded.

3.2 Tools and technologies used

Category	Technology
Programming	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Optional Deep Learning	TensorFlow / Keras
Database	MySQL / PostgreSQL
IoT Integration	ESP32 / compatible sensor gateway
Visualization	Power BI / Streamlit / Python
Development Environment	VS Code / Jupyter Notebook
Version Control	Git
Deployment	Cloud or local server

3.3 Solution overview

Data Sources Data Acquisition Cleaning and Validation Structured Storage Feature Engineering
 AI Prediction Model AQI/Air-Quality Analysis Dashboard and Alerts

3.4 Engineering standards applied

The design applies modularity, separation of concerns, documentation, version control, input validation, error handling and repeatable model evaluation. Data quality practices include missing-value detection, duplicate detection, timestamp validation, unit consistency checks and outlier review.

3.5 Ethical Standards applied

- Collect only data necessary for the system purpose.
- Protect location or user-related data where applicable.
- Clearly distinguish predictions from certified measurements.
- Document data limitations and model limitations.
- Avoid misleading alerts or unsupported health claims.
- Control access to stored data and system administration.
- Maintain human oversight for important decisions.

3.6 Solution Justification

The proposed solution combines monitoring and prediction in one architecture. A conventional display-only system explains what is happening now, while the AI component attempts to estimate what may happen next based on available data. The modular design also allows sensors, data sources and prediction models to be improved independently.

CHAPTER 4

RESULT AND RECOMMENDATIONS

4.1 Evaluation of results

The system should be evaluated using data quality, model accuracy and usability measures. Data processing can be assessed using completeness, validation success and processing reliability. Regression or forecasting models can be compared using MAE, RMSE and R^2 where appropriate. Lower prediction error is generally preferable, but evaluation must include comparison with a simple baseline such as persistence or historical average.

The expected result is a working analytical platform that transforms environmental measurements into understandable trends and future estimates. Actual numerical performance depends on the dataset, sensor quality, feature availability and testing procedure; this report does not claim a fixed accuracy without experimental evidence.

4.2 Challenges encountered

- Missing or inconsistent environmental measurements.
- Sensor noise and calibration limitations.
- Different sampling intervals across data sources.
- Limited historical data for model training.
- Changing weather and local events affecting predictions.
- Overfitting and poor generalization.
- Designing clear visualizations without oversimplifying the data.

4.3 Possible Improvements

- Add more real-time sensors and locations.
- Incorporate richer weather information.
- Use automated sensor-health checks.
- Apply advanced time-series or deep-learning models where justified.
- Add notification and alert services.
- Deploy the system on scalable cloud infrastructure.

4.4 Recommendations

Future implementation should prioritize reliable data before complex modelling. Sensor calibration, data validation and adequate historical coverage are more important than selecting an unnecessarily complicated algorithm. Prediction models should be tested periodically against new observations, and dashboard users should be informed about uncertainty and data freshness.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key learning outcomes

5.1.1 Academic Knowledge

The project improved understanding of environmental data analytics, machine learning, time-series concepts, data preprocessing and predictive evaluation. It also demonstrated how theoretical AI concepts can be connected to a practical engineering problem.

5.1.2 Technical Skills

- Python programming.
- Data cleaning and preprocessing.
- Feature engineering.
- Machine-learning model development.
- Database handling.
- IoT-oriented data acquisition concepts.
- Data visualization and dashboard development.
- Model evaluation and comparison.

5.1.3 Problem solving and critical thinking

The project required the larger problem to be divided into data acquisition, preprocessing, modelling and presentation stages. It also reinforced that an AI model should not be considered successful merely because it produces a prediction; its outputs must be tested against observed data and practical requirements.

5.2 Challenges encountered and overcome

The main technical challenge was connecting data quality with model quality. The solution approach emphasized cleaning and validation before training. Another challenge was selecting an appropriate model, which was addressed by comparing candidate methods rather than assuming that one algorithm would always perform best.

5.3 Applications of Engineering Standards

Engineering practices such as modular design, maintainability, documentation, testing, version control and error handling were applied to improve system reliability and future extensibility.

5.4 Applications of ethical standards

Ethical principles were considered through data minimization, responsible communication of predictions, security controls and avoidance of unsupported claims. The system should provide information for awareness and decision support rather than make uncontrolled decisions about people or communities.

5.5 Insights into the industry

The project demonstrates that real-world AI systems depend heavily on data engineering. A sophisticated algorithm cannot compensate for unreliable data. Environmental analytics therefore requires a combination of sensing, data management, machine learning, software engineering and responsible communication.

5.6 Conclusion of personal development

Overall, the project strengthened technical, analytical and engineering skills by requiring an end-to-end solution from raw data to user-facing results. It also highlighted the importance of validation, uncertainty, maintainability and ethical responsibility in AI-based systems.

CHAPTER 6

CONCLUSION

6.1 Summary of key findings

The AI-Driven Air Quality Monitoring and Prediction System provides an integrated approach to environmental data analysis. The proposed system collects data, validates and preprocesses measurements, applies AI models and presents results through a dashboard. The modular architecture supports future improvements without requiring the complete system to be redesigned.

6.2 Impacts and significance

- Improved visibility of historical and current air-quality patterns.
- Predictive insight based on available environmental data.
- Support for environmental awareness and planning.
- Practical application of AI and data analytics.
- A foundation for future IoT and cloud-based monitoring.

6.3 Future prospects

- Dense real-time sensor networks.
- Advanced spatio-temporal prediction models.
- Automated anomaly detection.
- Cloud-native deployment.
- Mobile applications and notifications.
- Geospatial heat maps.
- Continuous model monitoring and retraining.
- Integration with broader smart-city environmental systems.

CHAPTER 7

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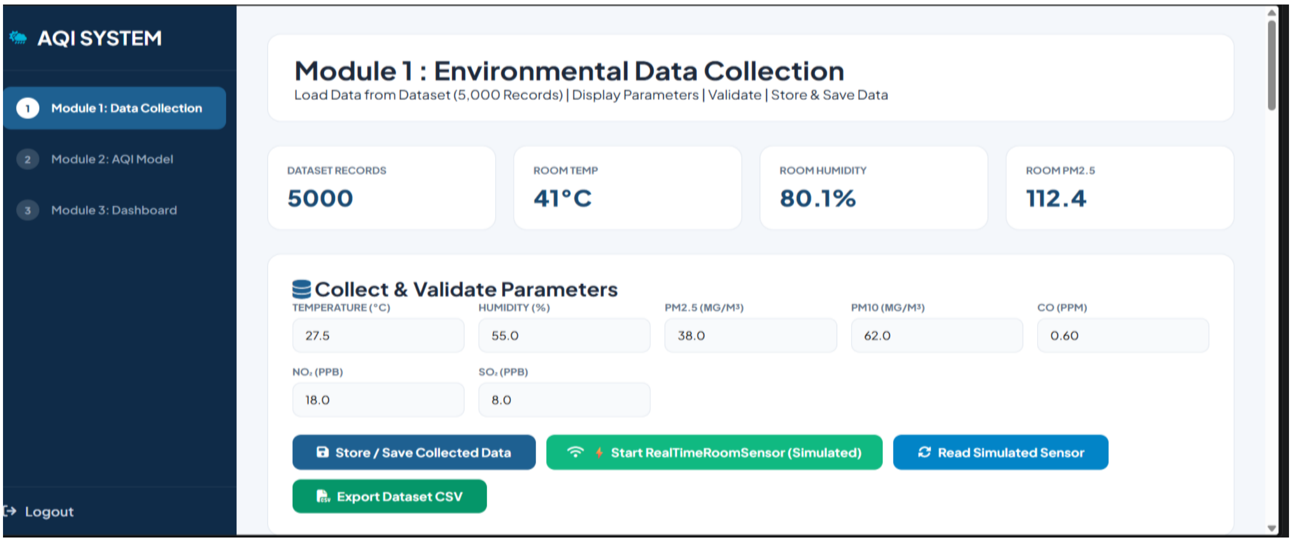
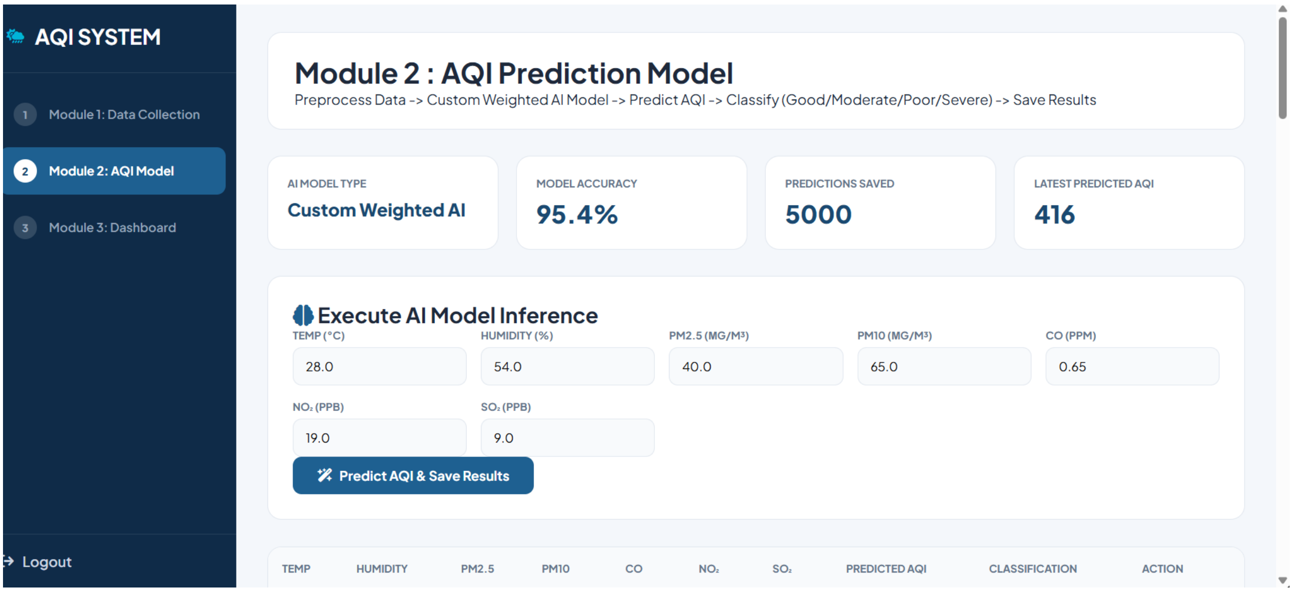
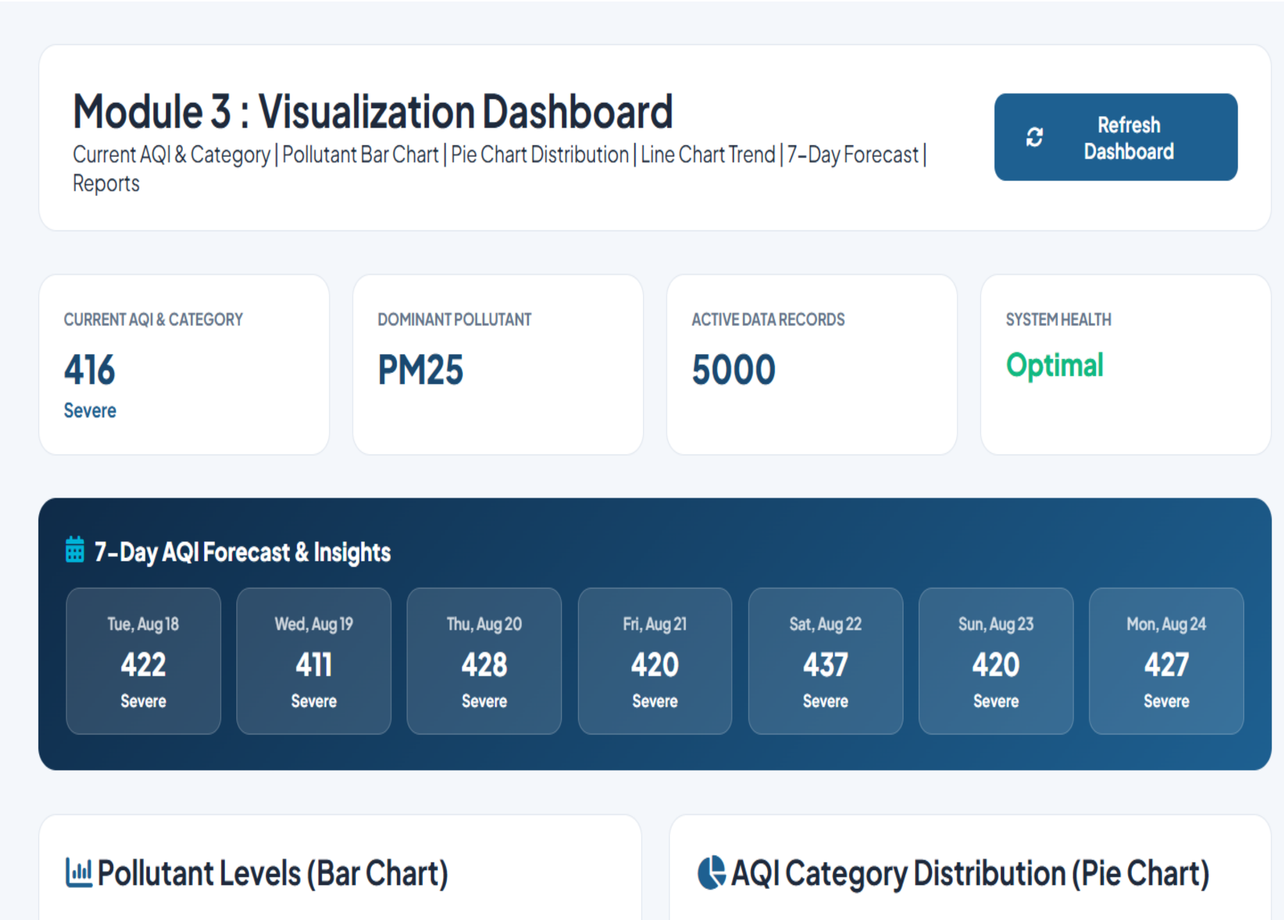
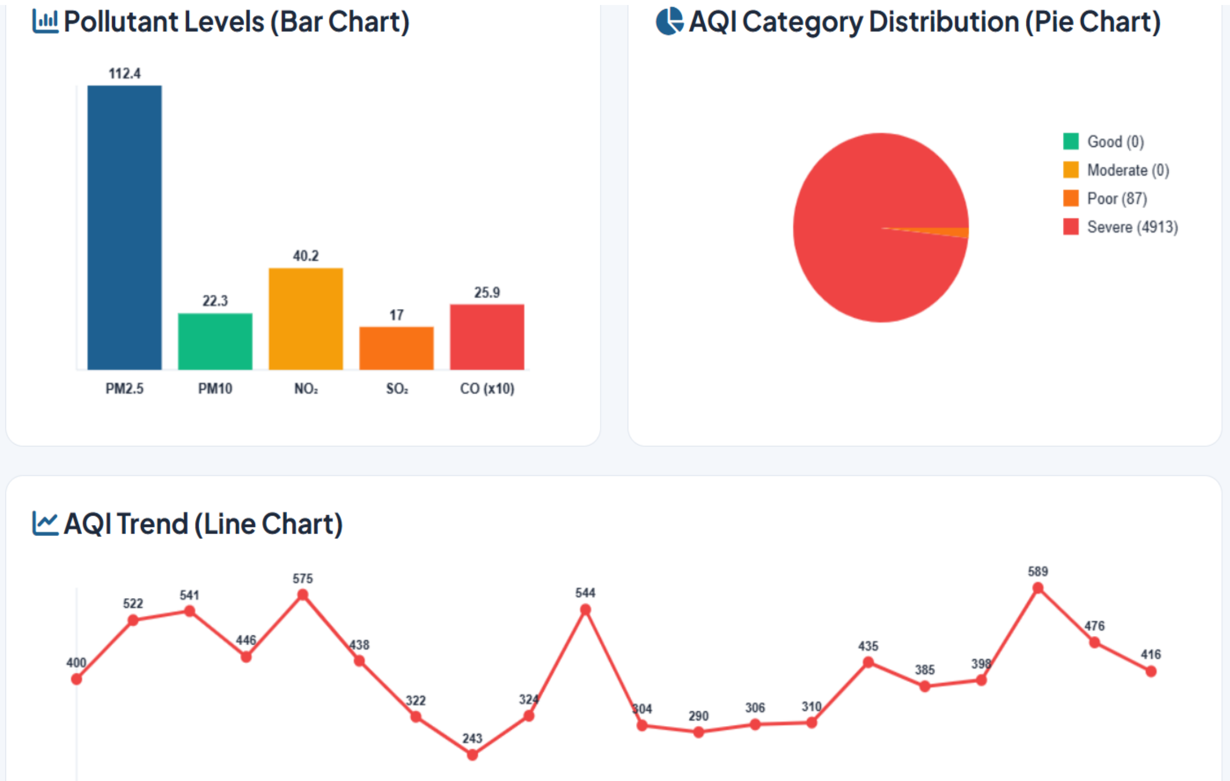
CHAPTER 8

APPENDICES






Module 3 : Visualization Dashboard
Current AQI & Category | Pollutant Bar Chart | Pie Chart Distribution | Line Chart Trend | 7-Day Forecast | Reports
Refresh Dashboard
CURRENT AQI & CATEGORY
416
Severe
DOMINANT POLLUTANT
PM25
ACTIVE DATA RECORDS
5000
SYSTEM HEALTH
Optimal
7-Day AQI Forecast & Insights
Tue, Aug 18
422
Severe
Wed, Aug 19
411
Severe
Thu, Aug 20
428
Severe
Fri, Aug 21
420
Severe
Sat, Aug 22
437
Severe
Sun, Aug 23
420
Severe
Mon, Aug 24
427
Severe
Pollutant Levels (Bar Chart)
AQI Category Distribution (Pie Chart)


Pollutant Levels (Bar Chart)
112.4
22.3
40.2
17
25.9
PM2.5
PM10
NO₂
SO₂
CO (x10)
AQI Category Distribution (Pie Chart)
Good (0)
Moderate (0)
Poor (87)
Severe (4913)
AQI Trend (Line Chart)
400
522
541
446
575
438
322
243
324
544
304
290
306
310
435
385
398
589
476
416